Results in Materials

A Conceptual Framework for a Rechargeable Hydride-Ion Battery Based on the Mo^{6+}/H^- Redox Couple

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Abstract:	Recent advances in anionic redox and hydride-ion conductors have opened new oppor- tunities for energy storage systems beyond conventional cationic intercalation. Here, a novel rechargeable battery concept is proposed, employing the Mo6+/Mo4+ redox couple as the cathodic host and the hydride ion (H–) as the mobile anion. Unlike established Li+ and Na+ intercalation mechanisms in MoO3, this design exploits the high reducing potential of H– and the structural tolerance of layered molybdenum oxides to hydrogen insertion, forming a reversible hydrogen bronze phase HxMoO3. Based on established density functional theory studies reported in the literature and redox potential analysis Mo6+/H– system could achieve higher theoretical energy density while utilizing globally abundant, non-toxic elements. The proposed chemistry represents a sustainable and cost-effective alternative to alkali-ion systems and provides a framework for future exploration of hydride-based solid-state batteries.

Dear Editor,

Please find enclosed our manuscript entitled **"A Rechargeable Hydride-Ion Analog to Alkali-Metal Batteries Based on the Mo■■/H■ Redox Couple"**, which we respectfully submit for consideration as a research article in your journal.

This work proposes a new conceptual framework for rechargeable hydride-ion energy storage based on the reversible intercalation of hydride ions into a molybdenum oxide host. Unlike conventional alkali-metal battery chemistries that rely on Li■, Na■, or K■ charge carriers, the proposed system explores hydride ions (H■) as the active ionic species and the reversible Mo■■/Mo■■ redox couple as the electrochemically active center.

The manuscript presents the electrochemical concept, theoretical capacity and energy-density estimates, and a critical evaluation of the proposed chemistry in the context of existing literature on hydride-ion conductors, hydrogen bronzes, and transition-metal oxide intercalation. We discuss the scientific rationale, anticipated advantages, and key challenges associated with this battery chemistry while identifying future computational and experimental studies required to evaluate its practical feasibility.

We believe the manuscript will be of interest to researchers working in energy storage, electrochemistry, solid-state ionics, and materials science because it introduces a potentially new direction for rechargeable battery research based on abundant elements and an unconventional charge carrier.

This manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have approved the manuscript and agree with its submission.

Thank you for your consideration. We appreciate your time and look forward to your response.

Sincerely,

Asif Kazi

A Rechargeable Hydride-Ion Analog to Alkali-Metal Batteries: Mo6+/H– System as a High-Potential Reversible Energy Storage Concept

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1 Abstract

Recent advances in anionic redox and hydride-ion conductors have opened new opportunities for energy storage systems beyond conventional cationic intercalation. Here, a novel rechargeable battery concept is proposed, employing the Mo^{6+}/Mo^{4+} redox couple as the cathodic host and the hydride ion (H^-) as the mobile anion. Unlike established Li^+ and Na^+ intercalation mechanisms in MoO_3, this design exploits the high reducing potential of H^- and the structural tolerance of layered molybdenum oxides to hydrogen insertion, forming a reversible hydrogen bronze phase H_xMoO_3. Based on established density functional theory studies reported in the literature and redox potential analysis Mo^{6+}/H^- system could achieve higher theoretical energy density while utilizing globally abundant, non-toxic elements. The proposed chemistry represents a sustainable and cost-effective alternative to alkali-ion systems and provides a framework for future exploration of hydride-based solid-state batteries.

2 Introduction

The global demand for energy storage technologies that exceed the performance, safety, and sustainability limits of Li-ion systems has intensified research into alternative charge carriers. Among the promising directions are multivalent cations (Mg^{2+}, Al^{3+}), and recently, anionic systems such as O^{2-} and H^- have emerged as viable candidates [1] [2]

Hydride-ion batteries (HIBs) leverage the high redox potential of H^- and its low mass to enable high gravimetric energy densities. However, most known hydride systems suffer from irreversible hydrogen evolution, unstable metal-hydride interfaces, or lack of compatible host frameworks [3]. In this context, molybdenum oxides (MoO_3) present a compelling host candidate due to their layered orthorhombic structure, multi-electron redox flexibility ($Mo^{6+}/Mo^{5+}/Mo^{4+}$), and proven stability in Li^+ and Na^+ intercalation batteries.

This work proposes a rechargeable Mo^{6+}/H^- system as an anionic analog to conventional alkali-ion batteries, exploiting the reversible formation of hydrogen bronzes (H_xMoO_3) as the fundamental energy storage mechanism.

3 Theoretical Framework

3.1 Redox Chemistry

In conventional Li-ion systems:

$$\mathrm{MoO_3} + x\mathrm{Li^+} + xe^- \leftrightarrow \mathrm{Li}_x\mathrm{MoO_3} \tag{1}$$

For the hydride analog:

$$\mathrm{MoO_3} + x\mathrm{H^-} \leftrightarrow \mathrm{H}_x\mathrm{MoO_3} + xe^- \tag{2}$$

Here, Mo^{6+} is reduced through intermediate valence states upon hydride insertion. The process can be reversed electrochemically if a solid hydride-conducting electrolyte is employed.

3.2 Energy Density and Potential

The standard reduction potential of H^- formation from H_2 is approximately $-2.25\,\mathrm{V}$ vs. SHE, significantly lower than that of Li/Li^+ ($-3.04\,\mathrm{V}$). When paired with a MoO_3 cathode ($E^\circ \approx +0.5$ V vs. SHE), the theoretical cell potential approaches 2.7–3.0 V, comparable to commercial Li-ion systems, but using lighter and more abundant elements.

3.3 Energy Density

The gravimetric capacity of MoO_3 can be calculated using:

$$C_{\mathrm{th}} = \frac{nF}{3.6M} \tag{3}$$

where $n = 2$ electrons per Mo atom, $F = 96485$ C/mol, and $M = 143.95$ g/mol is the molar mass of MoO_3. Substituting the values:

$$C_{\mathrm{th}} = \frac{2 \times 96485}{3.6 \times 143.95} \approx 372\ \mathrm{mAh\ g^{-1}}$$

Assuming an average cell voltage $V \approx 2.8$ V for the Mo^{6+}/H^- system, the theoretical gravimetric energy density is:

$$E_{\text{th}} = V \cdot C_{\text{th}} = 2.8 \times 372 \text{ mAh g}^{-1} \approx 1041.6 \text{ mWh g}^{-1} \approx 1042 \text{ Wh kg}^{-1}$$

This value represents the theoretical limit based on the active cathode material only. Practical full-cell energy densities are expected in the range of 300–500 Wh kg^{-1}, considering electrolyte, current collectors, and packaging.

4 Materials and Abundance

Molybdenum is a relatively abundant transition metal, with major reserves in China, the United States, Chile, and Peru. Global molybdenum production exceeds 250,000 tonnes annually - byproduct of copper mining. Hydrogen is measurably inexhaustible and its storage as hydride is ubiquitous. Thus, both elemental components of the proposed system are selectively abundant but globally distributed, ensuring long-term scalability and low resource risk given geopolitical stability.

5 Experimental Feasibility

Pre-existing studies show that hydrogen can be inserted into MoO_3 to form H_xMoO_3 with tunable optical and electronic properties [4]. Additionally, hydride-ion-conducting solids such as BaH_2, CaH_2, and oxyhydrides (e.g., $LaH_{3\text{-}x}O_x$) provide the necessary electrolytic options for reversible ion transport. The key challenges involve suppressing parasitic H_2 evolution and stabilizing the H^- conduction pathway under cycling.

6 Discussion

The Mo^{6+}/H^- system represents a paradigm shift towards anion-based energy storage. Its primary advantage lies in reversing the traditional charge carrier direction: electrons move opposite to hydride ions, enhancing electrochemical symmetry and potentially reducing polarization losses.

The reversibility of the hydrogen bronze phase and the availability of stable hydride-ion conductors together suggest that the Mo^{6+}/H^- pair could serve as the foundation for a new class of rechargeable batteries. If experimentally verified, this chemistry could deliver high capacity, long cycle life, and environmental sustainability unmatched by current Li-ion technology.

7 Conclusion

A conceptual framework for a rechargeable hydride-ion battery using MoO_3 as a redox-active host has been exhibited. The Mo^{6+}/H^- pair promises high theoretical energy density, low material cost, and abundant elemental availability. Future development may focus on experimental synthesis of H_xMoO_3 thin films, interface engineering with solid hydride electrolytes, and in-situ spectroscopy to confirm reversible hydride intercalation. This proposal outlines a new direction for green, abundant, and high-performance energy storage materials which may be a lucrative alternative to currently used lithium ion batteries.

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References

[1] G. Kobayashi, Y. Hinuma, S. Matsuoka, A. Watanabe, M. Iqbal, M. Hirayama, M. Yonemura, T. Kamiyama, I. Tanaka, R. Kanno, Pure h$^-$ conduction in oxyhydrides, Science 351 (6279) (2016) 1314–1317. doi:10.1126/science.aac9185.

[2] L. Hu, K. Sun, J. Peng, Y. Zhao, W. Zhang, Molybdenum oxide-based materials for lithium-ion batteries: A review, Nano Energy 31 (2017) 49–67. doi:10.1016/j.nanoen.2016.11.007.

[3] Y. Kobayashi, Y. Tsujimoto, H. Kageyama, Property engineering in perovskites via modification of anion chemistry, Annual Review of Materials Research 48 (2018) 303–326. doi:10.1146/annurev-matsci-070317-124415.

[4] M. Nagai, K. Tanaka, Y. Sasaki, Hydrogen insertion into molybdenum trioxide and formation of hydrogen bronzes h_xmoo_3, Journal of Solid State Chemistry 18 (5) (1976) 327–334. doi:10.1016/0022-4596(76)90225-0.

Declaration of interests

☒The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

☐The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: